

Year 12 Mathematics - Trial HSC 2004

QUESTION 1

- | | MARKS |
|--|-------|
| (a) Find the value of $e^{-2.5}$ correct to 3 significant figures. | 2 |
| (b) Factorise fully: $16x^2 - 36y^2$. | 2 |
| (c) Solve for t : $\frac{4}{2t-3} = \frac{5}{t}$. | 3 |
| (d) If $\frac{12}{2+\sqrt{10}}$ is written in the form $m + n\sqrt{10}$, where m and n are rational numbers, find the values of m and n . | 3 |
| (f) A customer is given a 6% discount on the purchase of a radio. If the customer paid \$42.30, find the price of the radio before the discount. | 2 |

QUESTION 2: (START A NEW PAGE)

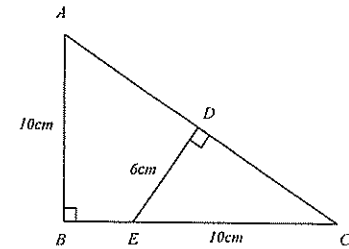
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|---|---|
| (a) Differentiate the following with respect to x , leaving your answer in simplest form. | |
| (i) $(3-4x)^7$. | 2 |
| (ii) $\frac{2x}{3x+1}$. | 2 |
| (b) (i) Find: $\int \frac{6}{1-2x} dx$. | 2 |
| (ii) Evaluate: $\int_0^{\frac{\pi}{2}} \sec^2 3x dx$. | 3 |
| (c) Find the equation of the curve $y = f(x)$, if $f'(x) = \frac{\sqrt{x}-4}{x}$ and the curve passes through the point $(1, 5)$. | 3 |

QUESTION 3: (START A NEW PAGE)

- | | MARKS |
|--|-------|
| (a) (i) Sketch the graph of $y = 3 \cos 2\theta$ for $0 \leq \theta \leq \pi$. | 2 |
| (ii) Solve $3 \cos 2\theta = 1$ for $0 \leq \theta \leq \pi$. Give your answer correct to 2 decimal places. | 2 |
| (b) (i) On the same set of coordinate axes, sketch the functions $y = 6x - x^2$ and $y = 2x$, clearly showing the coordinates of their intersection points. | 4 |
| (ii) Find the area bounded by the above curves and the x -axis. | 4 |

QUESTION 4: (START A NEW PAGE)

- (a) Triangles ABC and CDE are right angled at B and D respectively (as shown in the diagram).



- | | |
|--|---|
| (i) Copy the diagram onto your examination answer sheet and prove that $\triangle ABC$ and $\triangle CDE$ are similar. | 2 |
| (ii) If $AB = EC = 10\text{cm}$ and $DE = 6\text{cm}$, find the length of AC . | 2 |
| (b) $A(5, 20)$, $B(30, 15)$, $C(20, -10)$ and D are the vertices of a quadrilateral $ABCD$. | |
| (i) Given that the diagonals AC and BD are perpendicular, prove that the point D lies on the line $y = \frac{1}{2}x$. | 2 |
| (ii) If also $AB = AD$, prove that the coordinates of D are $(-6, -3)$. | 3 |
| (iii) Prove that AC bisects BD . | 3 |

QUESTION 5: (START A NEW PAGE)

- | | MARKS |
|---|-------|
| (a) A balloon drifts 100km from point A to point B on a bearing of 028°T . At point B the balloon changes direction and drifts 160km to point C on a bearing of 114°T . | |
| (i) Draw a neat diagram showing the above information. | 1 |
| (ii) Find the distance from point A to point C . Give your answer correct to the nearest kilometre. | 2 |
| (iii) Find the true bearing of point C from point A . Give your answer correct to the nearest degree. | 3 |
| (b) Water flows into then out of a container at a rate (R litres/minute) given by $R = t(10 - t)$. | |
| (i) Find the maximum flow rate. | 2 |
| (ii) Find an expression for the volume, V litres, of water in the container at time t minutes assuming that the container is initially empty. | 2 |
| (iii) Find the total time for the container to fill and then empty. | 2 |

QUESTION 6: (START A NEW PAGE)

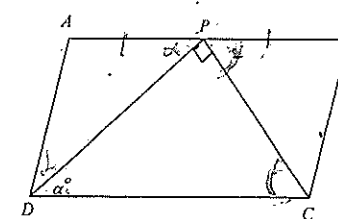
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|--|---|
| (a) (i) Sketch the region bounded by the curve $y = \sqrt[3]{x}$, the y -axis and the line $y = 2$. | 1 |
| (ii) Find the <i>exact</i> volume of the solid formed when the area in part (i) is rotated one revolution about the x -axis. | 4 |
| (b) The velocity v m/s of an object at time t seconds is given by $v = 3t^2 - 14t + 8$. The object is initially 30m to the right of the origin. | |
| (i) Find the initial acceleration of the object. | 1 |
| (ii) Find when the object is at rest. | 2 |
| (iii) Find the minimum distance between the origin and the object during its motion. | 4 |

QUESTION 7: (START A NEW PAGE)

- | | MARKS |
|---|-------|
| (a) On an interval $x_1 \leq x \leq x_2$, a curve $y = f(x)$ has the following three properties:
$f(x_1) < 0$, $f'(x) > 0$ and $f''(x) < 0$.
Draw a section of the curve $y = f(x)$ that illustrates all of above information. | 3 |
| (b) The mass M grams of a radioactive isotope of Carbon (called Carbon 14 and written as C_{14}) found in a rock sample at time t years is given by the formula $M = Ae^{-kt}$, where A and k are constants. | |
| (i) Prove that the rate of decay of the mass of C_{14} is proportional to the mass present at any time t . | 2 |
| (ii) If there is initially 100 grams of C_{14} and this mass decays to 75 grams in 2500 years, find the values of the A and k . Give your value of k correct to three significant figures. | 3 |
| (iii) Find the amount of C_{14} present at the end of 4000 years. Give your answer correct to the nearest gram. | 2 |
| (iv) Find the time required for the mass of C_{14} to decay to 5 grams. Give your answer correct to the nearest 100 years. | 2 |

QUESTION 8: (START A NEW PAGE)

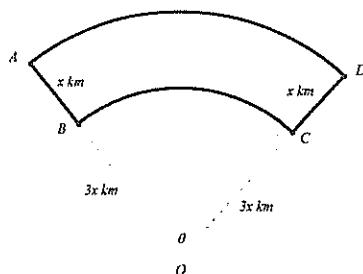
- | | |
|--|---|
| (a) As wire is unwound from a cylinder, the mass of wire remaining on the cylinder decreases. It is given that the mass, M kg, of wire remaining after t minutes can be calculated by the formula $M = 240 - 40\sqrt{t+1}$. | |
| (i) Find the initial mass of wire on the cylinder. | 1 |
| (ii) Find the time taken to remove all the wire from the cylinder. | 2 |
| (iii) Find the rate at which the wire is being removed from the cylinder when half the wire has been removed. | 3 |
| (b) $ABCD$ is a parallelogram. P is a point chosen on side AB so that PD bisects $\angle ADC$ and $\angle DPC = 90^\circ$. (as shown in the diagram) | |
| (i) If $\angle PDC = \alpha^\circ$, prove that $\angle BPC = (90 - \alpha)^\circ$. | 3 |
| (ii) Prove that $\triangle BPC$ is isosceles. | 3 |



QUESTION 9: (START A NEW PAGE)

MARKS

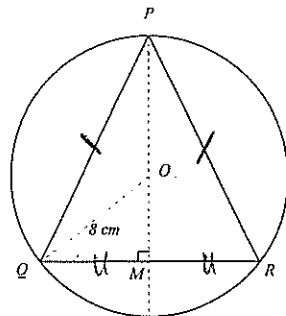
- (a) Four towns A, B, C and D are joined by roads that are either straight or arcs of concentric circles with centre at O . Towns B and C are distance $3x$ km from O and towns A and D are both distance x km from B and C respectively and $\angle AOD = \theta$ radians. (see diagram)



- (i) Write an expression, in terms of x and θ , for the length of the journey from town A to town D along the arc AD . 1
- (ii) A salesperson wants to travel from town A to town D but must visit towns B and C on the way. Write an expression, in terms of x and θ , for the length of this journey from town A to town D . 1
- (iii) Find the value of θ for which the journeys described in parts (i) and (ii) are the same distance. 2

- (b) An isosceles triangle PQR with $PQ = PR$ is inscribed in a circle of radius 8 cm (as shown in the diagram).

Given that O is the centre of the circle and M is the midpoint of the base QR of the triangle, you may assume that P, O and M are collinear and PM is perpendicular to QR .



- (i) If the height, PM cm, of $\triangle PQR$ is h cm, prove that its area, A cm², is given by $A = h\sqrt{16h - h^2}$. 3
- (ii) Write down the restriction on the values for h . 1
- (iii) Find the maximum area of $\triangle PQR$. 4

QUESTION 10: (START A NEW PAGE)

MARKS

A fund is established to provide prizes for a basketball team's annual Awards night. \$10 000 is placed in the fund one year before the first Awards night. It is decided that \$450 will be withdrawn from the fund each year to purchase the annual prizes. The money in the fund is invested at 3% p.a. compounded annually with the interest paid into the fund before each annual Awards night.

- (i) Show that the fund contains \$9695.50 after the second Awards night. 2
- (ii) If A_n is the amount in dollars remaining in the fund after the n^{th} Awards night, prove that $A_n = 5000(3 - 1.03^n)$. 3
- (iii) Find the amount of money in the fund after the 25th Awards night. Give your answer correct to the nearest dollar. 1
- (iv) Find the maximum number of Awards nights that can be financed using this fund. 2
- (v) For the fund described above it is decided to increase the amount of money withdrawn for each Awards night by 2% each year.
- (α) Show that the amount remaining in the fund after the 2nd Awards night is \$9686.50. 2
- (β) Find the amount remaining in the fund after the 25th Awards night. Give your answer correct to the nearest dollar. 2



THE END



Year 12 Mathematics - Trial HSC 2004

QUESTION 1

(a) $e^{-2.5} = 0.0821$ (to 3 significant figures)

(b) $16x^2 - 36y^2 = 4(4x^2 - 9y^2)$
 $= 4(2x - 3y)(2x + 3y)$

(c) $4t = 5(2t - 3)$
 $= 10t - 15$
 $6t = 15$
 $t = 2\frac{1}{2}$

(d) $\frac{12}{2 + \sqrt{10}} = \frac{12(2 - \sqrt{10})}{(2 + \sqrt{10})(2 - \sqrt{10})}$
 $= \frac{12(2 - \sqrt{10})}{4 - 10}$
 $= \frac{12(2 - \sqrt{10})}{-6}$
 $= -2(2 - \sqrt{10})$
 $= -4 + 2\sqrt{10}$
 $\therefore m = -4, n = 2$

(f) 94% of radio price = \$42.30
1% of radio price = $\frac{\$42.30}{94}$
100% of radio price = $\frac{\$42.30}{94} \times 100$
 $= \$45.00$

QUESTION 2: (STAR A NEW PAGE)

(a) (i) Let $f(x) = (3 - 4x)^7$ 2
 $f'(x) = 7(3 - 4x)^6 \times (-4)$
 $= -28(3 - 4x)^6$

(ii) Let $f(x) = \frac{2x}{3x+1}$ 2
 $f'(x) = \frac{(3x+1)(2) - (2x)(3)}{(3x+1)^2}$
 $= \frac{6x+2-6x}{(3x+1)^2}$
 $= \frac{2}{(3x+1)^2}$

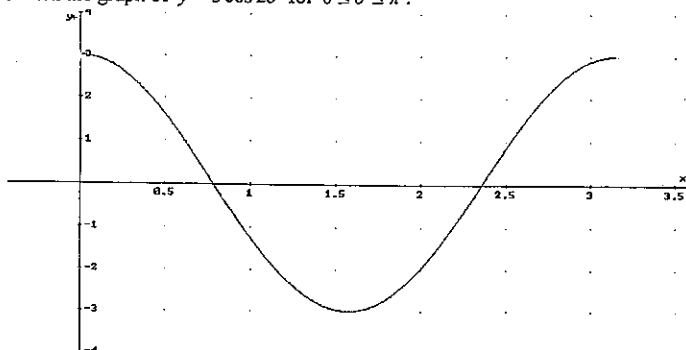
(b) (i) $\int \frac{6}{1-2x} dx = -3 \ln(1-2x) + c$ 2

(ii) $\int_0^{\frac{\pi}{4}} \sec^2 3x dx = \frac{1}{3} [\tan 3x]_0^{\frac{\pi}{4}}$ 3
 $= \frac{1}{3} \{ \tan \frac{3\pi}{4} - \tan 0 \}$
 $= -\frac{1}{3}$

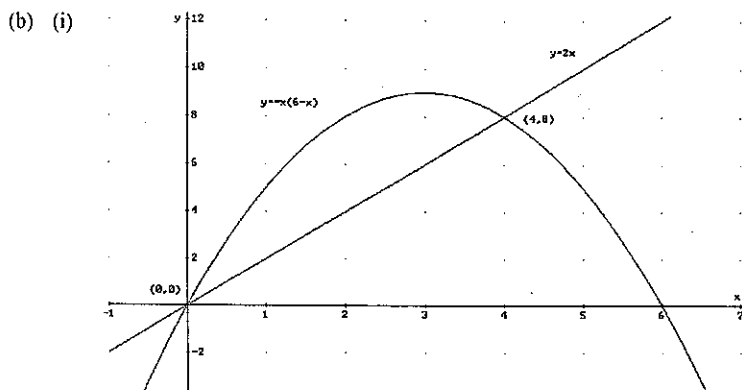
(c) $\frac{dy}{dx} = \frac{\sqrt{x} - 4}{x}$ 3
 $= x^{-\frac{1}{2}} - \frac{4}{x}$
 $y = 2x^{\frac{1}{2}} - 4 \ln x + c$
at point (1,5)
 $5 = 2\sqrt{1} - 4 \ln 1 + c$
 $\therefore c = 3$
 $y = 2\sqrt{x} - 4 \ln x + 3$

QUESTION 3: (STAR A NEW PAGE)

- (a) (i) Sketch the graph of $y = 3 \cos 2\theta$ for $0 \leq \theta \leq \pi$.



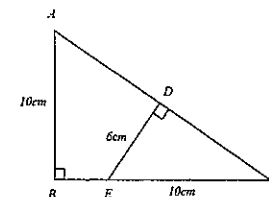
- (ii) $3 \cos 2\theta = 1$
 $\cos 2\theta = \frac{1}{3}$
 $2\theta = 1.230959$ or 5.052226
 $\theta = 0.62$ or 2.53 (to 2 decimal places)



- (ii)
- $$A = \int_0^4 2x \, dx + \int_4^6 (6x - x^2) \, dx$$
- $$= \left[x^2 \right]_0^4 + \left[3x^2 - \frac{1}{3}x^3 \right]_4^6$$
- $$= (4^2 - 0) + \left((3 \times 6^2 - \frac{1}{3} \times 6^3) - (3 \times 4^2 - \frac{1}{3} \times 4^3) \right)$$
- $$= 25\frac{1}{3}$$
- Area = $25\frac{1}{3} \text{ u}^2$

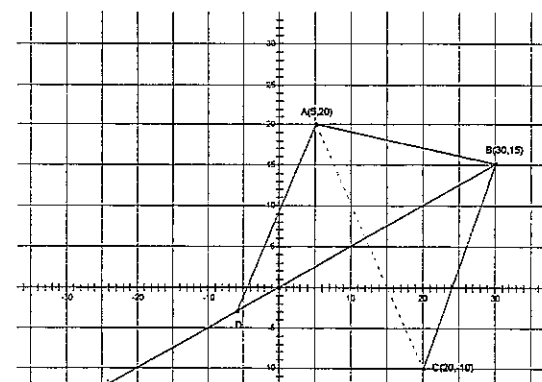
QUESTION 4: (STAR A NEW PAGE)

- (a) (i) In $\triangle ACB$ and $\triangle CDE$
 $\hat{ACB} = \hat{DCE}$ (common)
 $\hat{ABC} = \hat{CDE}$ (both 90°)
 $\triangle ABC \equiv \triangle EDC$ (equiangular)



- (ii) $\frac{AC}{10} = \frac{10}{6}$ (ratio of corresponding sides in similar triangles)
 $AC = 16\frac{2}{3}$
length of $AC = 16\frac{2}{3} \text{ cm}$

- (b) (i)
- $$\text{slope } AC = \frac{20+10}{5-20} = -2$$
- $$\therefore \text{slope } DB = \frac{1}{2}$$
- equation DB is
- $$y - 15 = \frac{1}{2}(x - 30)$$
- $$y - 15 = \frac{1}{2}x - 15$$
- $$y = \frac{1}{2}x$$



- (ii) Let D be the point $(2a, a)$
 $AD^2 = AB^2$
 $(2a - 5)^2 + (a - 20)^2 = (5 - 30)^2 + (20 - 15)^2$
 $5a^2 - 60a - 225 = 0$
 $a^2 - 12a - 45 = 0$
 $(a + 3)(a - 15) = 0$
 $a = -3$ or 15
at point D , $a = -3$
 $\therefore D$ is $(2a, a) = (-6, -3)$

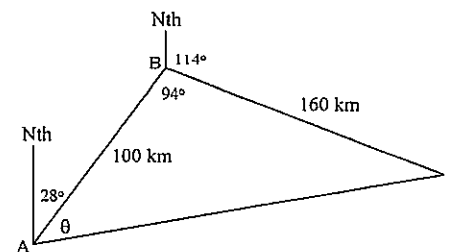
(iii) Using coordinate geometry
 Midpoint of BD is $P(12,6)$
 Equation of line AC is
 $y - 20 = -2(x - 5)$
 $y = -2x + 30$
 at point $P(12,6)$
 $LHS = y$
 $= 6$
 $RHS = -2x + 30$
 $= -2 \times (-12) + 30$
 $= 6$
 $\therefore LHS = RHS$
 $\therefore$ midpoint P lies on line AC
 i.e. line AC bisects DB

or Using congruent triangles
 Let AC meet BD at P
 In $\triangle ADP$ and $\triangle ABP$
 $AD = AB$ (given)
 $AP = AP$ (common)
 $\angle APD = \angle APB$ (both 90° , $AC \perp BD$)
 $\therefore \triangle ADP \cong \triangle ABP$ (RHS)
 $\therefore DP = BP$ (corresponding sides in congruent triangles)
 $\therefore$ line AC bisects DB

3

QUESTION 5: (STAR A NEW PAGE)

(a) (i)



$$(ii) AC^2 = 100^2 + 160^2 - 2(100)(160)\cos 94^\circ$$

$$AC = 195 \text{ km (to nearest km)}$$

$$(iii) \cos \theta = \frac{100^2 + AC^2 - 160^2}{2(100)(AC)}$$

$$\approx 0.5715$$

$$\theta = 55^\circ 8'$$

$$\text{bearing} = (28^\circ + 55^\circ)T$$

$$= 083^\circ T \text{ (to nearest degree)}$$

(b) (i) when $t = 5$, $R = 5(10 - 5)$
 $= 25$
 max. flow rate = 25 L/min

(ii) $V = \int (10t - t^2) dt$
 $= 5t^2 - \frac{1}{3}t^3 + c$
 when $t = 0$, $V = 0 \Rightarrow c = 0$
 $V = 5t^2 - \frac{1}{3}t^3$

(iii) when $V = 0$
 $5t^2 - \frac{1}{3}t^3 = 0$
 $\frac{1}{3}t^2(15 - t) = 0$
 $t = 0$ or 15
 $\therefore$ time taken = 15 minutes

1

1

2

3

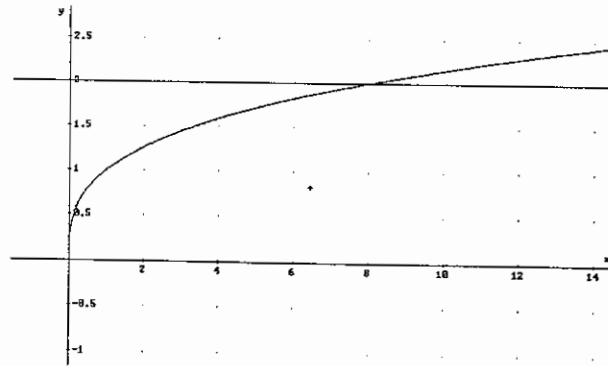
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2

2

QUESTION 6: (STAR A NEW PAGE)

(a) (i)



$$\begin{aligned}
 (ii) \quad V &= \pi \int_0^8 (2^2 - x^2) dx \\
 &= \pi \left[4x - \frac{1}{3}x^3 \right]_0^8 \\
 &= \pi \left(4 \times 8 - \frac{1}{3} \times 8^3 \right) - (0) \\
 &= \frac{64\pi}{3} \\
 \therefore \text{volume} &= \frac{64\pi}{3} \text{ u}^3
 \end{aligned}$$

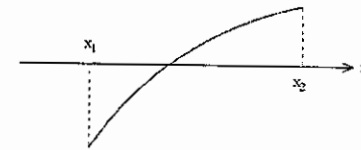
$$\begin{aligned}
 (b) \quad (i) \quad v &= 3t^2 - 14t + 8 \\
 a &= 6t - 14 \\
 \text{when } t &= 0, a = -14 \\
 \text{initial acceleration} &= -14 \text{ ms}^{-2}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \text{at rest when } v &= 0 \\
 3t^2 - 14t + 8 &= 0 \\
 (3t - 2)(t - 4) &= 0 \\
 t &= \frac{2}{3} \text{ or } 4 \\
 \text{at rest after } \frac{2}{3} \text{ seconds or } 4 \text{ seconds}
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad x &= t^3 - 7t^2 + 8t + c \\
 t = 0, x = 30 &\Rightarrow c = 30 \\
 \therefore x &= t^3 - 7t^2 + 8t + 30 \\
 \text{when } t &= 0, x = 30 \\
 \text{when } t &= \frac{2}{3}, \ddot{x} = -10 < 0, \therefore \text{concave down} \Rightarrow \text{local max. tp} \\
 \text{when } t &= 4, \ddot{x} = 10 > 0, \therefore \text{concave up} \Rightarrow \text{local min. tp.} \\
 x &= 4^3 - 7 \times 4^2 + 8 \times 4 + 30 \\
 &= 14 \\
 \therefore \text{minimum distance} &= 14 \text{ km}
 \end{aligned}$$

QUESTION 7: (STAR A NEW PAGE)

(a)



OR



$$\begin{aligned}
 (b) \quad (i) \quad \frac{dM}{dt} &= -kAe^{-kt} \\
 &= -kM \text{ since } M = Ae^{-kt} \\
 \therefore \frac{dM}{dt} &\propto M
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \text{when } t &= 0, M = 100 \\
 100 &= Ae^0 \\
 A &= 100 \\
 \text{when } t &= 2500, M = 75 \\
 75 &= 100e^{-2500k} \\
 e^{-2500k} &= 0.75 \\
 -2500k &= \ln 0.75 \\
 k &= \frac{\ln 0.75}{-2500} \\
 &= 1.15 \times 10^{-4} \text{ (to 3 significant figures)}
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad \text{when } t &= 4000 \\
 M &= 100e^{-4000 \times 1.15 \times 10^{-4}} \\
 &\approx 63 \\
 \text{mass} &\approx 63 \text{ grams}
 \end{aligned}$$

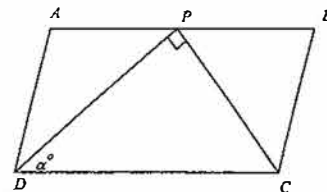
$$\begin{aligned}
 (iv) \quad \text{when } M &= 5 \\
 5 &= 100e^{-kt} \\
 e^{-kt} &= 0.05 \\
 -kt &= \ln 0.05 \\
 t &= \frac{\ln 0.05}{-k} \\
 &\approx 26049 \\
 \text{time} &= 26000 \text{ years (to nearest 100 years)}
 \end{aligned}$$

QUESTION 8: (STAR A NEW PAGE)

- (a) (i) when $t = 0$
 $M = 240 - 40\sqrt{t+1}$
 $= 200$
 $\therefore$ initial mass = 200kg
- (ii) when $M = 0$
 $0 = 240 - 40\sqrt{t+1}$
 $40\sqrt{t+1} = 240$
 $\sqrt{t+1} = 6$
 $t+1 = 36$
 $t = 35$
 $\therefore$ time = 35 minutes

(iii) $\frac{dM}{dt} = 0 - 40 \times \frac{1}{2}(t+1)^{-\frac{1}{2}}$
 $= -\frac{20}{\sqrt{t+1}}$
 when $M = 100$
 $100 = 240 - 40\sqrt{t+1}$
 $40\sqrt{t+1} = 140$
 $\sqrt{t+1} = 3.5$
 $\therefore \frac{dM}{dt} = -\frac{20}{3.5}$
 $= -\frac{40}{7}$
 $\therefore$ rate = $-\frac{40}{7}$ kg/min

- (b) (i) $AB \parallel DC$ (opposite sides of parallelogram are parallel)
 $\angle APD = \alpha^\circ$ ($AB \parallel DC$, alternate angles are equal)
 $\angle BPC + \alpha^\circ + 90^\circ = 180^\circ$ (straight angle) ($\angle APB = 180^\circ$)
 $\angle BPC = (90 - \alpha)^\circ$



- (ii) $\angle ADC = 2\alpha^\circ$ (PD bisects $\angle ADC$)
 $\angle ABC = 2\alpha^\circ$ (opposite angles of parallelogram are equal)
 $\angle BCP + 2\alpha^\circ + (90 - \alpha)^\circ = 180^\circ$ (angle sum of $\triangle BPC = 180^\circ$)
 $\angle BCP = (90 - \alpha)^\circ$
 $\therefore \triangle BPC$ is isosceles ($\angle BPC = \angle BCP = (90 - \alpha)^\circ$)

QUESTION 9: (STAR A NEW PAGE)

- (a) (i) $AD = 4x\theta$
- (ii) $ABCD = 2x + 3x\theta$
- (iii) $AD = ABCD$
 $4x\theta = 2x + 3x\theta$
 $x\theta - 2x = 0$
 $x(\theta - 2) = 0$
 $\theta = 2$ ($x \neq 0$)
 $\therefore$ angle = 2 radians

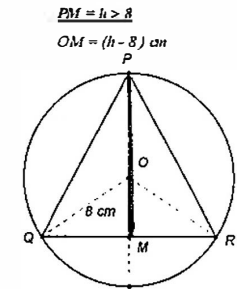
- (b) (i) Area = $\frac{1}{2}QR \cdot PM$
 $QR = 2QM$

Case (1)

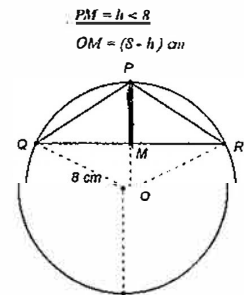
Case (1) if $h \geq 8$
 $QM^2 = 8^2 - (h-8)^2$ (Pythagoras' Theorem)

Case 2 if $h < 8$
 $QM^2 = 8^2 - (8-h)^2$ (Pythagoras' Theorem)

$\therefore QM^2 = 16h - h^2$
 $QM = \sqrt{16h - h^2}$ ($QM > 0$)
 $QR = 2\sqrt{16h - h^2}$
 Area: $A = \frac{1}{2} \times 2\sqrt{16h - h^2} \times h$
 $\therefore A = h\sqrt{16h - h^2}$
 $= 16h - h^2$
 $QM = \sqrt{16h - h^2}$ ($QM > 0$)
 $QR = 2\sqrt{16h - h^2}$
 Area: $A = \frac{1}{2} \times 2\sqrt{16h - h^2} \times h$
 $\therefore A = h\sqrt{16h - h^2}$



Case (2)



- (ii) $0 \leq h \leq 16$

(iii) $A = h(16h - h^2)^{\frac{1}{2}}$
 $\frac{dA}{dh} = (1)(16h - h^2)^{\frac{1}{2}} + (h) \times \frac{1}{2}(16h - h^2)^{-\frac{1}{2}} \times (16 - 2h)$
 $= \sqrt{16h - h^2} + \frac{8h - h^2}{\sqrt{16h - h^2}}$
 $= \frac{2h(12 - h)}{\sqrt{16h - h^2}}$

for stat. pt. $\frac{dA}{dh} = 0$

$$\frac{2h(12-h)}{\sqrt{16h-h^2}} = 0$$

$$2h(12-h) = 0$$

$$h = 0 \text{ or } 12$$

test stat. points

when $h = 0$, $A = 0$, $\therefore$ min.area

when $h = 12$

h	$<12 (=11)$	12	$>12 (=13)$
$\frac{dA}{dh}$	$\frac{2(11)(1)}{\sqrt{16 \times 11 - 11^2}}$	0	$\frac{2(13)(-1)}{\sqrt{16 \times 13 - 13^2}}$
	> 0		< 0

Change in gradient (+, 0, -) and curve is continuous for $11 \leq h \leq 13$ $\therefore$ stat. pt. is a local max. tp.

Since the area function is continuous for $0 \leq h \leq 16$ and there is only one max. tp. for $0 < h < 16$ then the local max. tp. is the absolute max.

$$\begin{aligned} \text{maximum } A &= 12\sqrt{12(16-12)} \\ &= 48\sqrt{3} \end{aligned}$$

$$\therefore \text{ maximum area} = 48\sqrt{3} \text{ cm}^2$$

QUESTION 10: (STAR A NEW PAGE)

- (i) Let $\$A_n$ = amount in the fund after the n^{th} awards night

2

$$A_1 = 10000 \times 1.03 - 450$$

$$A_2 = A_1 \times 1.03 - 450$$

$$= (10000 \times 1.03 - 450) \times 1.03 - 450$$

$$= 10000 \times 1.03^2 - (1.03 + 1) \times 450$$

$$= 9695.50$$

$$\therefore \text{ amount in fund} = \$9695.50$$

- (ii) $A_1 = 10000 \times 1.03 - 450$

3

$$A_2 = A_1 \times 1.03 - 450$$

$$= (10000 \times 1.03 - 450) \times 1.03 - 450$$

$$= 10000 \times 1.03^2 - (1.03 + 1) \times 450$$

$\vdots$

$$A_n = 10000 \times 1.03^n - (1.03^{n-1} + 1.03^{n-2} + \dots + 1.03 + 1) \times 450$$

$$= 10000 \times 1.03^n - 1 \times \left(\frac{1.03^n - 1}{1.03 - 1} \right) \times 450$$

$$= 10000 \times 1.03^n - \left(\frac{1.03^n - 1}{0.03} \right) \times 450$$

$$= 10000 \times 1.03^n - (1.03^n - 1) \times 15000$$

$$= 10000 \times 1.03^n - 1.03^n \times 15000 + 15000$$

$$= 15000 - 5000 \times 1.03^n$$

$$A_n = 5000(3 - 1.03^n)$$

- (iii) when $n = 25$

1

$$A_{25} = 5000(3 - 1.03^{25})$$

$$= 4531.11$$

$$\therefore \text{ amount} = \$4531 \text{ (to nearest dollar)}$$

- (iv) $A_n \geq 0$

2

$$5000(3 - 1.03^n) \geq 0$$

$$3 - 1.03^n \geq 0$$

$$1.03^n \leq 3$$

$$n \ln(1.03) \leq \ln 3$$

$$n \leq \frac{\ln(1.03)}{\ln 3}$$

$$n \leq 37.16$$

$$\therefore \text{ max. number of awards nights} = 37$$

(v) Let $\$B_n$ = amount in the fund after the n^{th} awards night 2

(α)

$$B_1 = 10000 \times 1.03 - 450$$

$$B_2 = B_1 \times 1.03 - 450 \times 1.02$$

$$= (10000 \times 1.03 - 450) \times 1.03 - 450 \times 1.02$$

$$= 10000 \times 1.03^2 - (1.03 + 1.02) \times 450$$

$$= 9686.5$$

$\therefore$ amount in fund = \$9686.50

(β) Let $\$B_n$ = amount in the fund after the n^{th} awards night 2

$$B_1 = 10000 \times 1.03 - 450$$

$$B_2 = B_1 \times 1.03 - 450 \times 1.02$$

$$= (10000 \times 1.03 - 450) \times 1.03 - 450 \times 1.02$$

$$= 10000 \times 1.03^2 - (1.03 + 1.02) \times 450$$

$$\vdots$$

$$B_n = 10000 \times 1.03^n - 450 \times \{1.03^{n-1} + 1.03^{n-2}(1.02) + 1.03^{n-3}(1.02^2) + \dots + 1.03(1.02^{n-2}) + 1.02^{n-1}\}$$

B_n series is a GP with $a = 1.02^{n-1}$ and $r = \frac{1.03}{1.02}$

$$B_n = 10000 \times 1.03^n - 450 \times 1.02^{n-1} \times \left\{ \frac{\left(\frac{1.03}{1.02} \right)^n - 1}{\left(\frac{1.03}{1.02} \right) - 1} \right\}$$

$$= 10000 \times 1.03^n - \frac{450 \times 1.02^{n-1} \left\{ \frac{1.03^n - 1.02^n}{1.02^n} \right\}}{\left(\frac{1.03 - 1.02}{1.02} \right)}$$

$$= 10000 \times 1.03^n - \frac{450 \times (1.03^n - 1.02^n)}{0.01}$$

$$= 10000 \times 1.03^n - 45000 \times (1.03^n - 1.02^n)$$

$$= 10000 \times 1.03^n - 45000 \times 1.03^n + 45000 \times 1.02^n$$

$$B_n = 45000 \times 1.02^n - 35000 \times 1.03^n$$

$$B_{25} = 45000 \times 1.02^{25} - 35000 \times 1.03^{25}$$

$\therefore$ amount in fund = \$545



THE END

